

RTutor.ai Quick-Start Guide

Two-Way Tables, Column Percents, Column Percent Bar Graphs, and Line Graphs
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0) Setup

This guide uses the Grade Interaction dataset.

1. Upload the dataset in RTutor.ai under **(1. Data)** → **Browse**.
2. Confirm four columns: *Year-in-School*, *Major*, *Format*, *Grade*.
3. Under **(2. Data Types)** → **Edit**, set Year-in-School, Major, and Format = Category, and Grade = Integer (or Numeric).

1) Two-Way Table

Goal: Create a two-way frequency table crossing Year-in-School (rows) and Format (columns).

Your prompt to RTutor.ai must:

1. Reference the Grade Interaction dataset and both variables: Year-in-School and Format.
2. Ask for a two-way table (also called a contingency table or cross-tabulation) with Year-in-School as the rows and Format as the columns.
3. Request row and column totals (marginals) so you can verify the counts.

Check: Does the table have two rows (Freshman, Sophomore) and two columns (In-Person, Online)? Do all cells have counts and do the totals add up?

2) Column Percents

Goal: Convert the two-way table into column percentages (each column sums to 100%).

1. Ask RTutor.ai to compute column percentages for the two-way table of Year-in-School by Format. Each Format column should sum to 100%.
2. Request rounding to one or two decimal places and clear labeling of each cell as a percentage.

Check: Does each column (In-Person, Online) sum to approximately 100%? If RTutor.ai computed row percentages instead, explicitly say “column percentages, where each column sums to 100%.”

3) Column Percent Bar Graph

Goal: Create a stacked or side-by-side bar graph showing column percentages.

1. Ask RTutor.ai for a stacked bar graph (or side-by-side bar graph) of Year-in-School by Format using column percentages, so each Format bar totals 100%.
2. Set the y-axis label to “Percentage,” font size 14, a color palette of your choice, and place the legend at the bottom.

Check: Does each bar reach 100%? Can you visually compare the proportion of Freshman vs Sophomore within each Format?

4) Line Graph

Goal: Create a line graph of mean Grade by Format, with separate lines for each Year-in-School group.

1. Ask RTutor.ai to compute the mean Grade for each combination of Format and Year-in-School.
2. Ask for a line graph with x-axis = Format (In-Person, Online), y-axis = Mean Grade, and separate lines with markers for Freshman and Sophomore.

3. Request font size 14, a color palette of your choice, and the legend at the bottom.

Check: Are there two lines (one per Year-in-School) each connecting two points? Can you see which group benefits more from a given format?

5) Interpretation & Extension

1. Write 2–3 sentences: Using the column percent bar graph, describe whether the distribution of Year-in-School differs between In-Person and Online formats.
2. Write 1–2 sentences about the line graph: Do both groups show the same pattern across formats, or does one group perform differently depending on the format?
3. Repeat the workflow using a different pair of categorical variables (e.g., Major × Format) and write the same interpretations.